Summary-statistics Mendelian randomization and GATE score

PDCD1 protein and type 1 diabetes

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The challenge: linked SNPs and summary statistics

Most GWAS data exist only as summary statistics (one coefficient per SNP)

- SNPs near each other are correlated — *linkage disequilibrium* (LD)
- Conventional Mendelian randomization uses only a single exposure-associated SNP in each region
- We want to use all exposure associated SNPs in each region to construct one scalar instrument for each region: for this we need multivariate (joint) regression coefficients

Individual-level data are often unavailable or siloed across secure servers

$\Rightarrow$ Can we recover multivariate coefficients from summary statistics alone, using a reference panel for LD?

Recovering multivariate coefficients via LD matrix inversion

Let $\hat{\beta}_u$ be the vector of univariate (marginal) per-SNP coefficients and $\mathbf{C}_g$ the LD correlation matrix from a reference panel.

Key result:

$$\hat{\beta}_m = \mathbf{D}^{-1}\,\mathbf{C}_g^{-1}\,\mathbf{D}\,\hat{\beta}_u$$

where $\mathbf{D} = \text{diag}(\text{per-SNP SDs})$.

In plain language — three steps:

1. Scale each coefficient to *per-SD-genotype* units
2. Multiply by the **inverse** LD correlation matrix (de-correlate)
3. Scale back to *per-allele* units

Near-singular LD (many SNPs in high LD) is handled by a *truncated pseudoinverse* retaining the K largest eigenvalues of $\mathbf{C}_g$ (effective rank).

Scalar instruments: one number per locus

Each locus yields a *scalar instrument* Z_i — a weighted sum of an individual's genotypes across all SNPs in the locus:

$$Z_i = \frac{\sum_j G_{ij}\,\hat{\beta}_{m,j}}{\|\hat{\boldsymbol{\beta}}_m\|}$$

Instrument-exposure coefficient $\hat{\alpha} = \|\hat{\boldsymbol{\beta}}_m\|$

- Expected change in exposure per 1-unit change in Z
- Measures instrument *strength*: how much the locus shifts the protein level

Everything computed from summary statistics using the UKBB EUR LD matrix ($N = 362{,}063$ samples, 18 million variants, GRCh37)

Instrument-outcome coefficient $\hat{\gamma}$

$\hat{\gamma}$ = log-OR of disease per unit change in Z (for binary outcomes)

From summary statistics — no individual-level outcome data needed:

- Obtain per-SNP outcome z-scores from an outcome GWAS
- Apply the same LD inversion to get multivariate outcome coefficients
- Project onto the scalar instrument Z

$$\hat{\gamma} = \hat{\alpha} \cdot \frac{\sum_j \hat{\sigma}_j^2 \, \hat{\gamma}_{u,j} \, \hat{\beta}_{m,j}}{\widehat{\mathrm{Var}}(S)}$$

Alternative — direct individual-level regression: score each individual on $Z_i = \sum_j G_{ij} \hat{\beta}_{m,j}$, regress disease on Z_i

Both methods implemented; direct is the ground truth for validation.

Standard errors

For $\hat{\alpha}$ (instrument-exposure):

$$\mathrm{SE}(\hat{\alpha}) = \sqrt{\frac{\sigma_X^2 - \mathrm{Var}(S)}{N \cdot \mathrm{Var}(Z)}}$$

Exposure variance σ_X^2 estimated from per-SNP marginal SEs and sample size.

For $\hat{\gamma}$ (instrument-outcome, binary):

$$\mathrm{SE}(\hat{\gamma}) = \frac{1}{\sqrt{N_Y \cdot \mathrm{Var}(Z) \cdot p(1-p)}}$$

- Only needs: outcome N_Y, case fraction p, $\mathrm{Var}(Z)$ from reference panel
- If reference LD $\neq$ target LD, $\mathrm{Var}(Z)$ is mis-estimated $\Rightarrow$ biased SE

Both SEs computed without access to individual-level data.

GATE score: pooling instruments across loci

GATE (Genomewide Aggregated *Trans*- Effects) combines all *trans*-pQTL instruments into a single composite genetic score:

$$G_i = \sum_{k \in \text{trans}} \hat{\alpha}_k \, Z_{ik}$$

- Each *trans*-pQTL k contributes its genetically predicted exposure value
- Standardise to SD $= 1$, then regress disease on G^*
- Tests whether the *overall* genetic evidence supports a causal effect

Intuition: if $\hat{\alpha}_k > 0$ at many loci and T1D tracks the predicted protein level, the composite score should associate with T1D more strongly than any single instrument.

GATE coefficient from summary statistics

Using the score test and independence of trans-loci, the GATE z-statistic reduces to the **IVW z-statistic** with weights $(\hat{\alpha}_k^*)^2$:

$$z_{\mathrm{GATE}} = \sqrt{Np(1-p)} \cdot \frac{\sum_k (\hat{\alpha}_k^*)^2 \, \hat{\theta}_k}{\sqrt{\sum_k (\hat{\alpha}_k^*)^2}}$$

where $\hat{\alpha}_k^* = \hat{\alpha}_k \, \mathrm{sd}_{Z,k}$ (per-SD units) and $\hat{\theta}_k = \hat{\gamma}_k^* / \hat{\alpha}_k^*$ is the per-locus IV ratio.

SE of $\hat{\beta}_{\mathbf{GATE}}$ simplifies to $1/\sqrt{Np(1-p)}$ — equivalent to a single 1-df binary logistic regression.

Validated against direct individual-level GATE regression ($z_{\mathrm{summary}} = 12.2$, $z_{\mathrm{direct}} = 9.7$; discrepancy consistent with reference–target LD mismatch).

Application: PDCD1 protein and type 1 diabetes

Exposure

- PDCD1 (PD-1): immune checkpoint protein; regulates T-cell activity
- UK Biobank Olink proteomics, $N = 33{,}902$ EUR participants
- 39 pQTL loci (MAF $\geq 1\%$, $p < 10^{-6}$, HLA excluded)

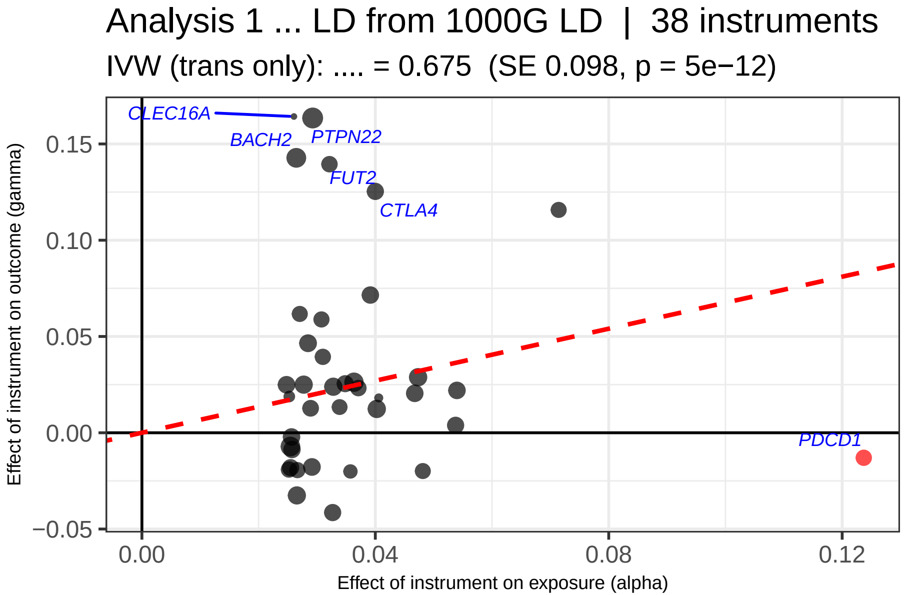
Outcome

- Type 1 diabetes: SDRNT1BIO cases + Generation Scotland controls
- 4,922 cases, 7,452 controls (all individual-level)

Three analyses

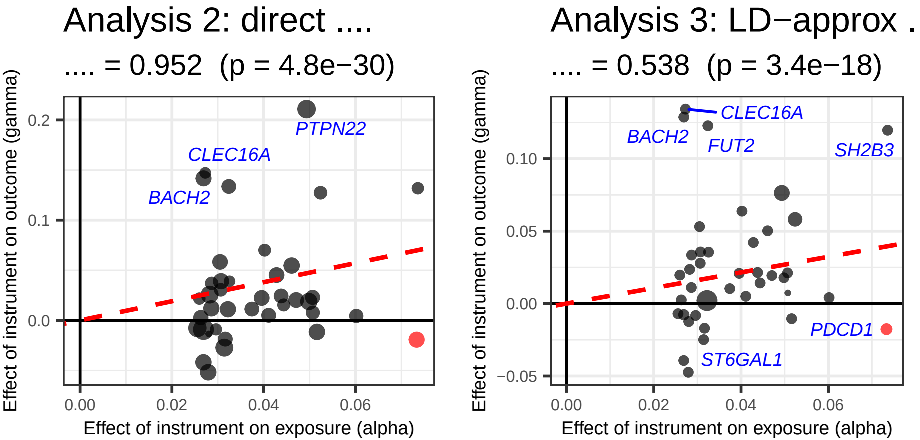
	$\hat{\alpha}$	$\hat{\gamma}$
Analysis 1 (Zhou et al.)	indiv. 1000G LD	indiv. 1000G LD
Analysis 2 (new)	sumstats UKBB LD	direct indiv.
Analysis 3 (new)	sumstats UKBB LD	sumstats UKBB LD

Analysis 1 (Zhou et al.): $\hat{\gamma}$ vs $\hat{\alpha}$



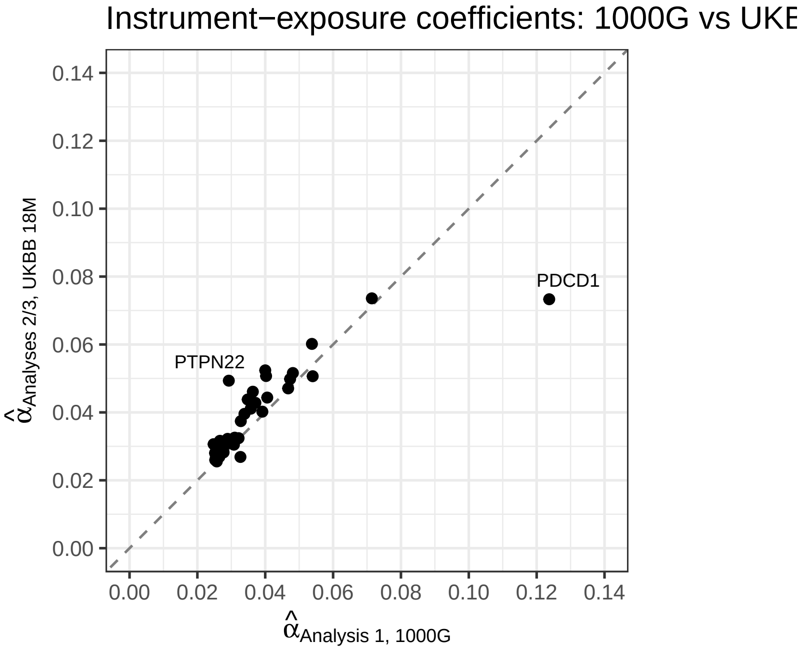
Red point: PDCD1 cis-pQTL. Red dashed line: IVW slope (trans only).

Analyses 2 and 3: LD matrix from UKBB 18M EUR

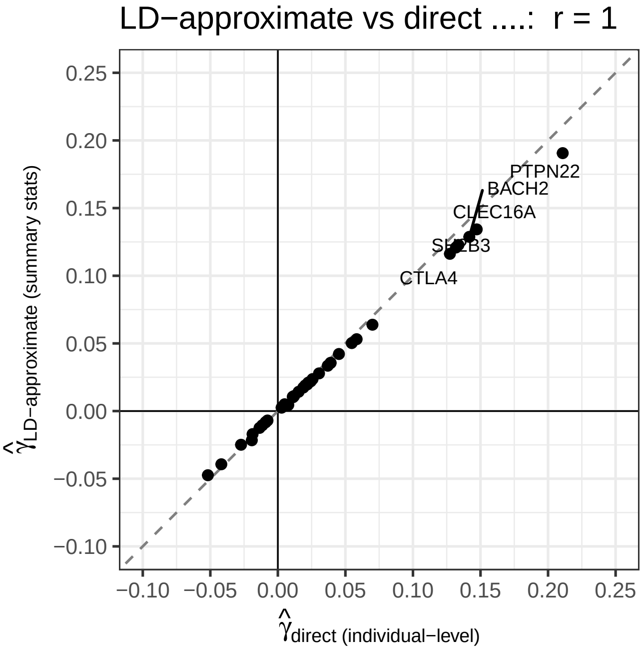


Both positive, both significant. UKBB LD reference is much larger (362k vs 503 samples).

Does the LD reference panel matter for $\hat{\alpha}$?



Does the LD-approximate $\hat{\gamma}$ agree with the direct estimate?



GATE score: summary-level vs direct regression

Method	$\hat{\beta}$	SE	z	p	Trans pQTLs
Summary-level (eqs. 7–8)	0.224	0.0184	12.22	2.5e-34	38
Direct regression	0.197	0.0202	9.73	2.3e-22	38

- Both methods: **strongly positive effect of PDCD1 on T1D**
- $z_{\text{direct}} = 9.7,\ p = 2.3e - 22$
- $z_{\text{summary}} / z_{\text{direct}} \approx 1.26$: reference panel slightly over-estimates instrument variance $\Rightarrow$ SE too small by $\sim 26\%$

Conclusions

Methods

- Multivariate SNP–exposure and SNP–outcome coefficients can be computed from summary statistics by inverting the LD correlation matrix
- SE of $\hat{\gamma}$ requires only N, case fraction, and reference-panel LD
- GATE z-statistic is equivalent to the IVW estimator restricted to trans-pQTLs; SE $= 1/\sqrt{Np(1-p)}$ when reference-panel LD is well-calibrated

PDCD1 / T1D results

- All three analyses consistently estimate a **positive causal effect** of circulating PDCD1 on T1D risk
- GATE score confirms this at genome-wide significance ($z \approx 9.7$ direct, $z \approx 12.2$ summary-level)

Validation

- Score tests agree with GLM ($r > 0.9999$, slope ≈ 1)